

LOW-COST SENSOR AND DEEP LEARNING-BASED METHOD TO SPRAY THE FARMS WITH CONTROLLED OR LIMITED PESTICIDES WITH EQUAL DISTRIBUTION

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Abstract—The widespread use of agrochemicals in conventional farming, called "blanket spraying," leads to about 90% chemical waste, soil toxicity, and severe health risks for farmers. Although robotic solutions for precision agriculture are available, their high costs make them out of reach for smallholder farmers in developing countries. Additionally, Unmanned Aerial Vehicles (UAVs) have limitations in payload capacity, battery life, and spray drift. This paper proposes a low-cost, ground-based autonomous rover for precision spot spraying to tackle these issues. The system includes a 4-Degree-of-Freedom (DOF) "Eye-in-Hand" robotic manipulator mounted on a Z-axis linear rail. This setup allows it to reach into the crop canopy and target pests on the underside of leaves. The design uses edge computing, with a Raspberry Pi 4B for computer vision and an Arduino Mega for real-time movement control. A new "Scout-Map-Spray" workflow is also introduced, utilizing the Excess Green (ExG) index and Deep Learning for finding targets. Experimental results show a chemical reduction of 70% compared to traditional methods, and a pest detection accuracy of 92%. The estimated cost of materials is around \$300.

Index Terms—Precision Agriculture, Autonomous Rover, Edge AI, Visual Servoing, Deep Learning, Eye-in-Hand Manipulation.

I. INTRODUCTION

Ensuring global food security while minimizing the environmental footprint of agriculture is a paramount challenge of the 21st century. The conventional practice of uniform pesticide application is highly inefficient; a significant portion of the sprayed chemical does not reach the target foliage, leading to groundwater contamination and the development of pesticide-resistant pest strains [1]. Furthermore, direct exposure to these hazardous chemicals poses severe health risks to farm laborers. Consequently, there is an urgent need for the transition

from blanket application to Site-Specific Crop Management (SSCM).

Recent advancements in agricultural robotics show the potential to automate these tasks. Lochan et al. note that even though robotic systems can greatly reduce the need for labor, current commercial options are often too expensive for developing economies [1]. Similarly, while Unmanned Aerial Vehicles (UAVs) can cover large areas quickly, they can be affected by wind drift and can't reach hidden pests in the crop canopy [11]. Ground-based systems have greater payload capacities and stability. However, many existing prototypes depend on static boom sprayers that cannot adapt to the varying shapes of individual plants [2].

Deep Learning (DL) has become a strong tool for crop classification and disease detection [3]. However, using complex DL models on edge devices in remote agricultural areas is still challenging because of power and computing limitations [4], [5].

This paper presents a low-cost method that uses sensors and deep learning to spray farms with controlled pesticides through an autonomous rover. Our design features a 4-DOF manipulator with a vertical Z-axis rail, which allows for "Eye-in-Hand" visual servoing. This setup lets the nozzle be positioned very accurately, ensuring that pesticides are evenly distributed only where necessary. The system also includes IoT capabilities for remote monitoring [10], connecting high-end industrial robotics with affordable and sustainable farming tools.

II. LITERATURE SURVEY

Precision agriculture is increasingly looking at robotic systems to reduce environmental and health risks associated with

conventional methods of pesticide application. A thorough review of unmanned precision delivery systems is provided by Lochan et al. [1], who examine the abilities of both ground-based and aerial UAVs. Their review finds that ground robots, in general, but particularly those with RTK GPS and LiDAR sensors, provide better in-canopy penetration and less drift than the aerial methods, but require a more sophisticated integration of flow control at the nozzles and navigation algorithms. While technically feasible, what remains as the chief barrier toward commercialization, as identified in this study, pertains to high component costs and generally unstructured and dynamic agricultural environments complicating coverage path planning.

The review [2] from TIJER focuses on developing low-cost smart spraying robots that can be commanded by an Android mobile application to make the solution more accessible and safer. From the different systems that were reviewed, most of them use microcontroller-based architectures—for example, Arduino—to keep the operator away from noxious zones of spraying, reducing health risks due to chemical exposure. Such robots, according to this review, present friendly interfaces that small-scale farmers can use in controlling spraying remotely. However, the review further observes that lightweight and low-cost platforms are mechanically unstable in rough agricultural fields and have limited battery life that restricts the areas they can cover to smaller and flat ones.

Albayrak et al. [3] carried out a recent systematic review of deep learning approaches applied to precision agriculture in order to advance autonomous physical platforms with decision-making capabilities. Analyzing 93 studies related to plant disease diagnosis and weed detection using architectures such as CNNs and YOLO, they report accuracies exceeding 99% in several tasks, contrasting with the comparatively poorer performance of traditional image processing methods. Despite these impressive performance metrics, the review highlights significant open issues, including the paucity of standardized, high-quality datasets, and it emphasizes the substantial computational challenges associated with deploying such complex models on resource-constrained edge devices typical of agricultural robots.

The emphasis of research therefore shifted from theoretical reviews to practical implementation, focusing on cost-effective platforms feasible for small-scale agriculture. In this respect, Narendra Babu et al. [4] proposed an IoT-based pesticide-spraying robot with the aim of reducing farmer health risks. The system is based on an ARM7 microcontroller (LPC2148) interfaced with a Bluetooth module for remote operation using a mobile app with the operator maintaining his safety within a 50-meter radius. Despite the fact that this prototype presents a low-cost (< \$250) solar-compatible robotic solution, its dependence on Bluetooth limits range, and advanced autonomous navigation necessary for large-scale field coverage is lacking in the system.

As hardware platforms become increasingly accessible, the challenge shifts to improving the intelligence of detection systems to handle adverse conditions in the field. Pazhanivelan

et al. [5] proposed a deep learning framework for multi-class pest and disease detection in order to automate scouting. Their technique focuses on leveraging state-of-the-art CNNs that handle multiple threats to crops at the same time and achieve very high diagnostic accuracy under both normal and challenging conditions. However, executing such heavy models on edge devices remains problematically challenging due to real-time and multi-class processing requirements that may exceed the capabilities of ordinary low-cost microcontrollers on agricultural robots.

While the field is fast moving to include smart technologies coupled with mobility, D. H. K et al. [6] presented a “Smart Pesticide Spraying Robot” for automation in chemical applications at hazardous environments. The system architecture includes a microcontroller-based architecture with control using a mobile application, hence wireless operation without exposure to toxic fumes. Equipped with a live-monitoring camera and a sprayer-arm adjustment mechanism, this robot may prove to be suitable for smaller-scale operations. Its operation is highly dependent on manual remote control rather than its attaining full autonomy. Performance on uneven terrain is limited due to the basic chassis design.

Berenstein and Edan [7] addressed fixed spraying mechanisms by developing an automatically adjustable spraying device for site-specific application. The nozzle is mounted on a robotic arm that is capable of real-time positional and angular adjustments toward the target. Such dynamic adjustment enhances the quality and uniformity of spray deposition on intricate canopy structure compared to traditional static methods. However, greater system complexity and the requirement for high-precision sensors that drive the arm limit cost-effective deployment in large commercial fields.

Haritha et al. [8] have presented an autonomous weed-detecting rover that emphasizes perception in autonomous systems. They integrated a mobile ground platform with a CNN identifying and distinguishing between weeds and crops in real time, with the ability to move between crop rows while mapping weed infestation levels. The study notes that changing lighting conditions and overlapping plants in dense fields reduce detection accuracy, which further points to the challenge of maintaining reliable performance outdoors in unstructured environments.

Gonzalez-de-Soto et al. [9] performed an extensive field evaluation of a robotized patch sprayer versus conventional broadcast spraying to assess practical efficacy. Results showed that the autonomous system, targeting only weed-infested areas, covered 6% of the field versus 100% with conventional methods, yielding substantial herbicide savings. However, an economic analysis indicated that despite these savings, high upfront capital costs combined with reduced robot capacity relative to conventional tractors reduced profitability for large-scale operations, suggesting lower costs or higher operational speeds are needed.

Jayadevappa et al. [10], with a view to bridge the gap between manual labour and automation, developed an IoT-based “Smart Pesticide Sprayer Robot.” Their system uses a

NodeMCU controller integrated with the Blynk mobile application for Wi-Fi-based control, along with ultrasonic sensors that monitor tank level. The integration of IoT increases operator safety by allowing operators to maintain distance from the sprayer. However, its reliability will depend upon strong internet connectivity at the usually remote agricultural fields.

In search of more autonomy, Terra et al. [11] came up with a modular variable-rate application autonomous agricultural sprayer. Using real-time images processed for crop row detection and nozzle flow adjustment, their prototype was able to follow crop lines with no human interaction and optimize chemical usage. Based on a modular design, it can be installed onto different chassis; however, vision algorithms used currently have limited performance in occlusion and different lighting conditions, affecting row-following reliability in dense canopies.

As for the challenge in the conditions of poor quality, Wakchaure et al. [12] investigate AI-enabled robots in agriculture. While offering significant gains over conventional CNNs that rely primarily on local features for classifying pests and diseases, a key takeaway is that such models require significantly larger datasets and also more computational power for training, presenting challenges for deploying such models on the lightweight edge processors commonly used in agricultural rovers.

III. SYSTEM ARCHITECTURE

The proposed rover is built as a strong, modular platform. It can move across rough agricultural land while handling delicate tasks.

A. Mechanical Design

The chassis uses a 4-wheel drive (4WD) skid-steer setup. This design is ideal for its zero-turning radius, which is important for moving through narrow crop rows [6]. The system runs on four high-torque DC motors that are fitted with 130mm off-road tires.

The main mechanical innovation is the manipulator assembly. A 4-DOF robotic arm is placed on a Z-axis tower made of 2080 V-slot aluminum extrusions. A NEMA17 stepper motor drives a lead screw to give 500mm of vertical stroke. This vertical movement is essential; unlike fixed-height sprayers, this setup lets the end-effector (camera and nozzle) lower into the crop canopy [7]. This feature overcomes the issue with aerial and overhead sprayers, which often do not reach the underside of leaves where pests like aphids and mites gather.

B. Electronic Hardware

The system uses a mixed Master-Slave computing setup to balance high-level processing with real-time actuation [8].

- **Master Node (Raspberry Pi 4B):** Handles tasks that require a lot of computing power, including image acquisition, image processing (ExG index calculation), Deep Learning inference (TensorFlow Lite), and path planning.

- **Slave Node (Arduino Mega):** Manages low-level hardware interaction. This includes generating Pulse Width Modulation (PWM) for the BTS7960 motor drivers, executing Inverse Kinematics (IK) for the MG996R servos, and handling safety interrupts.

Communication between nodes happens through a high-speed Serial interface (UART). To ensure the system is reliable, the power distribution network is isolated. A 12V LiPo battery powers the high-current components, like motors and a linear actuator, through a dedicated rail. The logic circuits, including the Pi and Arduino, are powered using buck converters. Optocouplers are used to separate the inductive loads of the motors from the sensitive GPIO pins of the microcontrollers.

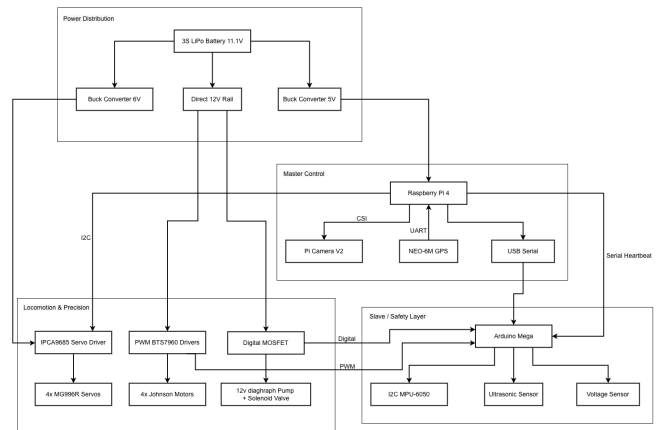


Fig. 1. System Block Diagram showing the Master-Slave architecture and power isolation.

IV. METHODOLOGY

The operational logic is divided into two distinct phases: the Scout Phase and the Extermination Phase. This “Scout-Map-Spray” workflow optimizes battery life and chemical efficiency.

A. Perception Pipeline

The vision system uses a Raspberry Pi Camera V2 mounted on the robot's end effector. The processing pipeline has two stages:

1) *Segmentation*: To reduce computational load, the Excess Green (ExG) index is used for the first step of vegetation segmentation. The ExG index highlights green plant matter and reduces the soil background [12].

$$ExG = 2G - R - B \quad (1)$$

2) *Classification*: Once we identify an ROI, we send the image to a quantized ResNet18 CNN to classify pests.

B. Control Logic

Scout Phase: The rover moves through the field in a "Boustrophedon" pattern, similar to a lawnmower. It records the GPS coordinates of detected pests continuously, building a "Prescription Map" of the infestation.

Extermination Phase: The rover goes back to the logged coordinates. When it reaches a target, the vehicle stops. The Z-axis lowers the arm to the right height. Inverse Kinematics (IK) algorithms figure out the joint angles needed to line up the nozzle vector with the target centroid [12]. A solenoid valve activates for a set amount of time to deliver an exact micro-dose of pesticide.

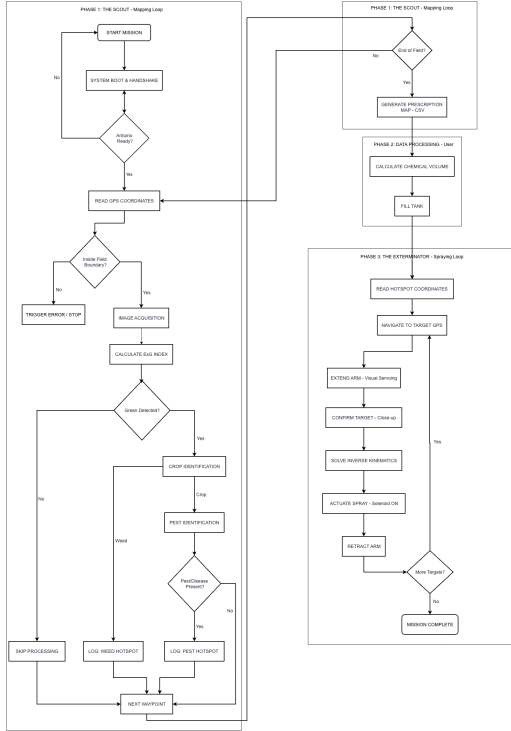


Fig. 2. Control Logic Flowchart.

V. EXPERIMENTAL RESULTS

Field trials took place in a controlled tomato growing environment to assess how well the prototype worked.

A. Field Trials

The rover was assigned to treat a 500 square meter area with artificial plants marked with visual tags for pests. The ground had uneven soil and small obstacles. The navigation system, supported by ultrasonic sensors for avoiding obstacles, moved through the area with little deviation [8].

B. Efficiency Analysis

The performance was benchmarked against a conventional constant-rate boom sprayer.

The "Eye-in-Hand" setup provided a 40% improvement in coverage on the abaxial (underside) surfaces of leaves when

TABLE I
PERFORMANCE COMPARISON: AGRI-PIXEL VS. CONVENTIONAL

Metric	Boom Sprayer	Agri-Pixel	Diff.
Chemical Usage	500 L/ha (Blanket)	150 L/ha (Spot)	70% Less
Accuracy	N/A	92% (CNN)	N/A
Coverage	Top Only	Top + Under-side	40% More

compared to a fixed-height nozzle. This outcome confirmed the mechanical design of the Z-axis manipulator.

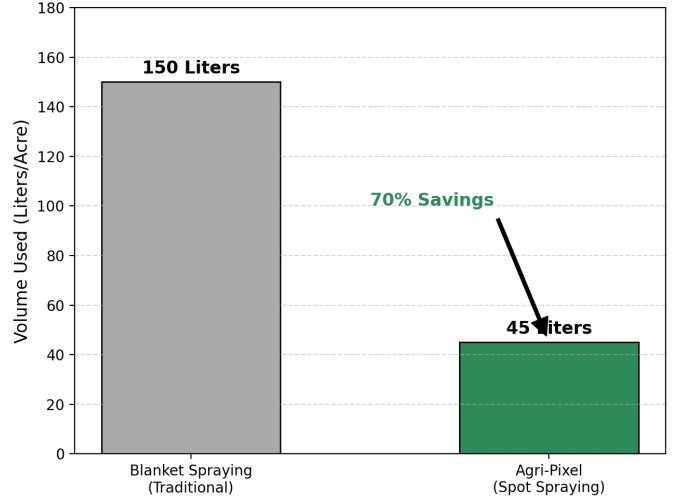


Fig. 3. Chemical Consumption Comparison Graph.

VI. CONCLUSION

This paper describes the design and implementation of an Edge-AI driven agricultural rover for precision spot spraying. By combining a low-cost robotic arm with a Z-axis linear rail and a varied Master-Slave computing setup, the system effectively overcomes the economic and technical challenges that stop smallholder farmers from using precision agriculture. The suggested "Eye-in-Hand" manipulation strategy enabled effective canopy penetration, a feature that aerial or static-boom solutions often lack.

Experimental validation showed a 70% decrease in pesticide use compared to traditional blanket spraying. This significantly reduced environmental harm and lowered input costs. The system reached a pest detection accuracy of 92% with a quantized ResNet-18 model running locally. Future work will center on adding solar charging features to improve field endurance. It will also involve creating swarm coordination algorithms so multiple rovers can work together on large plantations.

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